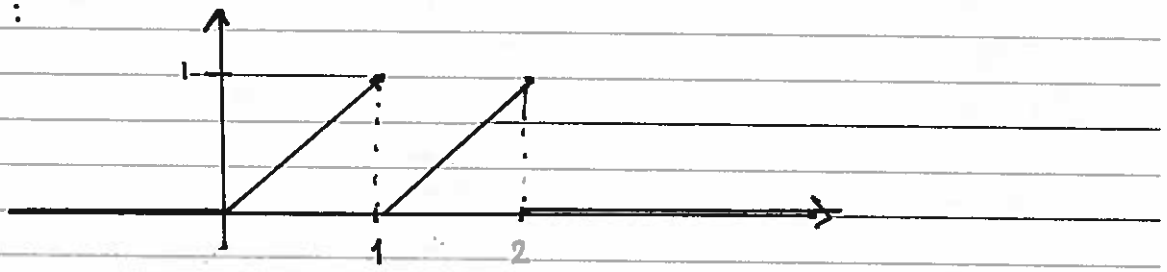
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1(a) Graph of f_X:



The function f_X is a density because ① $f_X(x) \geq 0$ for every $x \in \mathbb{R}$

② Area under the graph of f_X equals 1

i.e. $\int_{-\infty}^{\infty} f_X(x)\,dx = 1$ (NB. You can see this by inspection!).

(b) $\mathrm{Im}\,X = [0,2]$ because $f_X(x) > 0$ on $(0,2)$ and $f_X(x) = 0$ otherwise.

$$E(X) \underset{\text{(def'n)}}{=} \int_{-\infty}^{\infty} x f_X(x)\,dx = \int_0^2 x f_X(x)\,dx$$

$$= \int_0^1 x \cdot x\,dx + \int_1^2 x(x-1)\,dx = 7/6$$

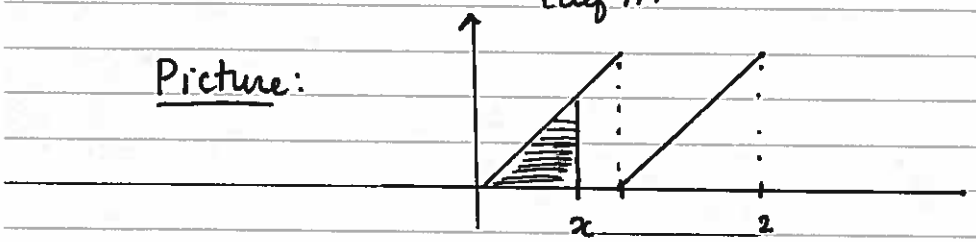
(c) Since $\mathrm{Im}\,X = [0,2]$, $F_X(x) = 0$ for $x < 0$

$$F_X(x) = 1 \text{ for } x \geq 2$$

For $0 \leq x < 1$, $F_X(x) = P(X \leq x) \underset{\text{(def'n)}}{=}$ Area under f_X up to x

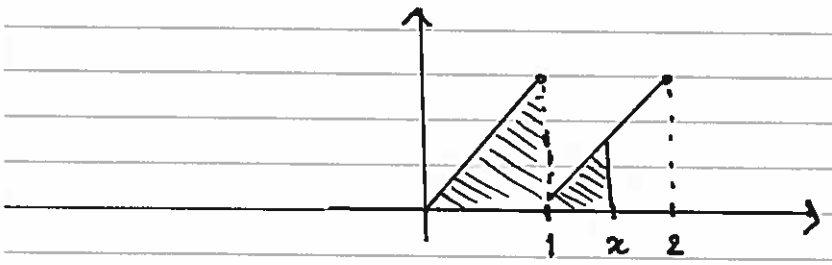
$= x^2/2$

Picture:



For $1 \leq x < 2$, $F_X(x) = P(X \leq x) =$ Area under f_X up to x

$$= \frac{1}{2} + \frac{(x-1)^2}{2}$$



Note For $1 \leq x < 2$, $f_X(x) = x - 1$ but $F_X(x) \neq \frac{x^2}{2} - x$

even though $\frac{d}{dx}\left(\frac{x^2}{2} - x\right) = x - 1 = f_X(x)$!!

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1(d)(i) $Y = X^2$. Since $\text{Im}\, X = [0,2]$, we have $0 \le X \le 2$.
Therefore, $0 \le X^2 \le 4$, i.e. $\text{Im}\, Y = [0,4]$.

For $0 \le y \le 4$, $F_Y(y) = P(Y \le y)$ by def'n of F_Y

$= P(X^2 \le y)$

$= P(-\sqrt{y} \le X \le \sqrt{y})$

$= P(X \le \sqrt{y}) - P(X < -\sqrt{y})$ ← This is 0 because X must be ≥ 0.

by def'n of F_X $= F_X(\sqrt{y})$

1(d)(ii) Since $\text{Im}\, Y = [0,4]$, $f_Y(y) = 0$ for $y \le 0$ or $y \ge 4$.

For $0 \le y \le 4$, $F_Y(y) = F_X(\sqrt{y})$

So: $f_Y(y) = F_Y'(y) = F_X'(\sqrt{y}) \cdot \frac{1}{2\sqrt{y}} = f_X(\sqrt{y}) \cdot \frac{1}{2\sqrt{y}}$

For $0 < y \le 1$, $f_X(\sqrt{y}) \cdot \frac{1}{2\sqrt{y}} = \sqrt{y} \cdot \frac{1}{2\sqrt{y}} = \frac{1}{2}$

For $1 < y \le 4$, $f_X(\sqrt{y}) \cdot \frac{1}{2\sqrt{y}} = (\sqrt{y} - 1) \cdot \frac{1}{2\sqrt{y}} = \frac{1}{2} - \frac{1}{2\sqrt{y}}$.

Putting this all together we obtain:

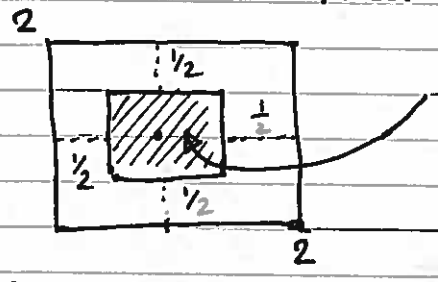
$$f_Y(y) = \begin{cases} \frac{1}{2} & 0 < y \le 1 \\ \frac{1}{2} - \frac{1}{2\sqrt{y}} & 1 < y \le 4 \\ 0 & \text{otherwise.} \end{cases}$$

Note: Some students found F_Y by computing $F_X(\sqrt{y})$ and then differentiated. This is also correct.

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2 (a) Since the random point Q is uniformly distributed in the square,

$$P(X > \tfrac{1}{2}) = \frac{\text{Area of shaded region}}{\text{Area of the square}} = \frac{1}{4}$$



NB $X > \tfrac{1}{2}$ if Q lands in the shaded square.
The shaded square has area = 1.

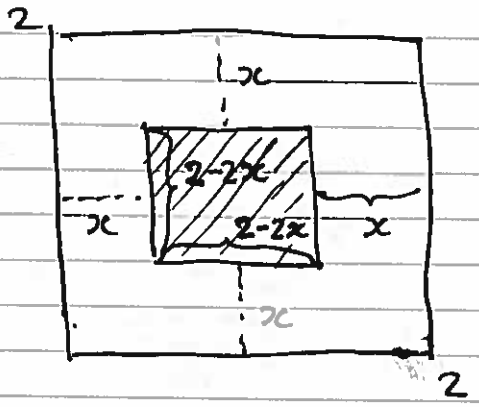
(b) Since the point Q can never be more than 1 unit from the boundary of the square, we must have $0 \le X \le 1$,

i.e. $\text{Im}\,X = [0,1]$. So $F_X(x) = 0$ for $x < 0$

$F_X(x) = 1$ for $x \ge 1$

For $0 \le x < 1$,

$$F_X(x) = P(X \le x) = 1 - P(X > x) = 1 - \frac{\text{Area shaded region}}{4}$$



$$= 1 - \frac{(2-2x)^2}{4}$$

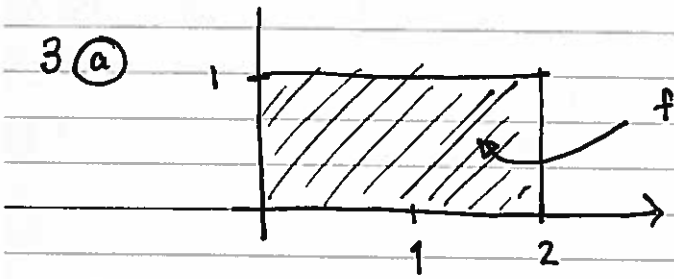
$$= 1 - (1-x)^2$$

(c) Density fn X is $f_X(x) = F_X'(x) = \begin{cases} 2-2x & 0 < x < 1 \\ 0 & \text{ow} \end{cases}$

So $E(X) = \int_0^1 x f_X(x)\,dx = \int_0^1 2x - 2x^2\,dx = \frac{1}{3}$.

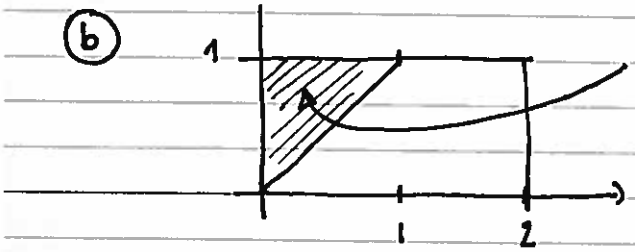
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3 (a)



$f_{X,Y}(x,y) > 0$ in the shaded region.

(b)



$X < Y$ in this region

So $P(X<Y) = \int_0^1 \int_0^y f_{X,Y}(x,y)\,dx\,dy = \int_0^1 \int_0^y \frac{1}{5}(2x+y)\,dx\,dy = \frac{2}{15}$

OR $P(X<Y) = \int_0^1 \int_x^1 f_{X,Y}(x,y)\,dy\,dx = \int_0^1 \int_x^1 \frac{1}{5}(2x+y)\,dy\,dx = \frac{2}{15}$

(c) Since the joint density $f_{X,Y}(x,y) = 0$ when $x<0$ or $x>2$ (see above)

we have $f_X(x) = 0$ for $x<0$ or $x>2$.

For $0 \le x \le 2$, $f_X(x) = \int_0^1 f_{X,Y}(x,y)\,dy = \int_0^1 \frac{1}{5}(2x+y)\,dy$

$= \frac{2x}{5} + \frac{1}{10}$.

(d) (i) $f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)} = \frac{\frac{1}{5}(2x+y)}{\frac{1}{5}(2x+\frac{1}{2})} = \frac{4x+2y}{4x+1}$ for $0 \le x \le 2$, $0 \le y \le 1$

and $f_{Y|X}(y|x) = 0$ otherwise.

(ii) $P(Y < \frac{1}{2} \mid X=1) = \int_0^{1/2} f_{Y|X}(y|1)\,dy = \int_0^{1/2} \frac{4+2y}{5}\,dy = \frac{9}{20}$

(iii) $E(Y|X=1) = \int_0^1 y\, f_{Y|X}(y|1)\,dy = \int_0^1 \frac{4y+2y^2}{5}\,dy = \frac{8}{15}$

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(4) (a) $X \sim Exp(1)$ and $Y \sim Exp(1)$ and X, Y are independent.

$Z = 3X + Y$, so $E(Z) = E(3X+Y) = 3E(X) + E(Y) = 4$
since $E(X) = E(Y) = 1$.

$Var(Z) = Var(3X+Y)$
$= Var(3X) + Var(Y)$ by the independence of X and Y
$= 9\,Var(X) + Var(Y)$
$= 10$ since $Var(X) = Var(Y) = 1$.

(b) Since X, Y are independent

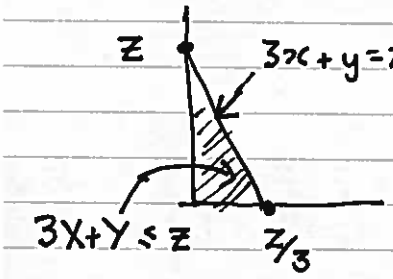
$$f_{X,Y}(x,y) = f_X(x) f_Y(y) = \begin{cases} e^{-x}e^{-y} & x, y > 0 \\ 0 & \text{otherwise.} \end{cases}$$

(c) Since $\text{Im}\, X = [0, \infty)$ and $\text{Im}\, Y = [0, \infty)$, $0 \le \underbrace{3X+Y}_{Z} < \infty$

i.e. $\text{Im}\, Z = [0, \infty)$ too

So $F_Z(z) = 0$ for $z < 0$

For $z \ge 0$, $F_Z(z) = P(Z \le z) = P(3X + Y \le z)$



$$= \int_0^{z/3} \int_0^{z-3x} f_{X,Y}(x,y)\, dy\, dx$$

$$= \int_0^{z/3} \int_0^{z-3x} e^{-x}e^{-y}\, dy\, dx$$

$$= \int_0^{z/3} -e^{-x-y} \Big|_0^{z-3x} dx$$

$$= \int_0^{z/3} e^{-x} - e^{-z+2x}\, dx$$

$$= -e^{-x} - \frac{e^{-z+2x}}{2} \Big|_0^{z/3} = 1 + e^{-z} - \frac{3e^{-z/3}}{2}$$

Alternatively,

For $z \ge 0$, $F_Z(z) = P(Z \le z) = P(3X + Y \le z)$

$$= \int_0^{z} \int_0^{\frac{z-y}{3}} f_{X,Y}(x,y)\, dx\, dy$$

$$= \int_0^{z} \int_0^{\frac{z-y}{3}} e^{-x}e^{-y}\, dx\, dy$$

$$= 1 + e^{-z} - \frac{3e^{-z/3}}{2}.$$

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5 (a) $F_M(x) = P(M \le x)$ by definition of F_M.

$= 1 - P(M > x)$

$= 1 - P(\min(X_1, X_2, \ldots, X_n) > x)$ by def'n of M

$= 1 - P(X_1 > x, X_2 > x, \ldots, X_n > x)$ since $\min(X_1, \ldots, X_n) > x$ iff $X_1 > x, X_2 > x, \ldots, X_n > x$.

$= 1 - P(X_1 > x)P(X_2 > x) \cdots P(X_n > x)$ by independence of $X_1, X_2, \ldots, X_n$

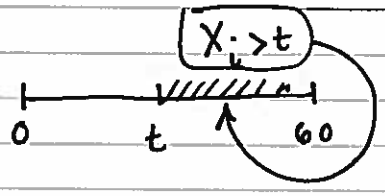
$= 1 - (1 - F_1(x)) \cdots (1 - F_n(x))$

since $P(X_K > x) = 1 - P(X_K \le x) = 1 - F_K(x)$.

(b) $T = \min(X_1, X_2)$, so $\text{Im}\, T = [0, 60]$ since $\text{Im}\, X_1 = [0, 60]$, $\text{Im}\, X_2 = [0, 60]$.

(i) So the density for T takes the value 0 when $x < 0$ or $x > 60$ i.e. $f_T(t) = 0$ for $t \le 0$ or $t \ge 60$.

For $0 < t < 60$, $F_T(t) = 1 - P(X_1 > t)P(X_2 > t)$ from part (a)



$= 1 - \left(\frac{60-t}{60}\right)\left(\frac{60-t}{60}\right)$ since $X_i \sim U(0, 60)$

$= 1 - \left(\frac{60-t}{60}\right)^2$

So $f_T(t) = F_T'(t) = \frac{d}{dt}\left(1 - \left(\frac{60-t}{60}\right)^2\right) = \frac{1}{30}\left(\frac{60-t}{60}\right)$.

(ii) Since the conditional distribution of Y given $T = t$ is $U(0, 60-t)$

The conditional expectation $E(Y|T=t) = \frac{60-t}{2}$ ← mean of $U(0, 60-t)$.

(iii) $E(W) = E(T+Y) = E(T) + E(Y)$

$E(T) = \int_0^{60} t f_T(t)\, dt = \int_0^{60} \frac{t(60-t)}{1800}\, dt = \frac{30t^2}{1800} - \frac{t^3}{5400}\Big|_0^{60} = 20$

$E(Y) = \int_0^{60} E(Y|T=t) f_T(t)\, dt$ by The Partition Theorem.

$= \int_0^{60} \left(\frac{60-t}{2}\right)\frac{(60-t)}{1800} = \frac{-(60-t)^3}{10800}\Big|_0^{60} = 20$

So $E(W) = E(T) + E(Y) = 20 + 20 = \underline{40 \text{ min}}$.

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6 (a) $M_Y(\theta) = E(e^{\theta Y})$ by definition.

$= E(e^{\theta(a+bX)})$ since $Y = a+bX$

$= E(e^{a\theta + b\theta X}) = E(e^{a\theta} \cdot e^{b\theta X})$ (algebra)

$= e^{a\theta} E(e^{b\theta X})$ by the properties of expected value.

$= e^{a\theta} M_X(b\theta)$ by definition of M_X.

(b) (i) If $W \sim N(3, 2)$ then using the formula sheet, we have $M_W(\theta) = e^{3\theta} e^{2\theta^2/2} = e^{3\theta} \cdot e^{\theta^2}$

$= \left(1 + 3\theta + \frac{(3\theta)^2}{2!} + \frac{(3\theta)^3}{3!} + \dots\right)\left(1 + \theta^2 + \frac{(\theta^2)^2}{2!} + \dots\right)$

by using the series expansion for e^x.

Since $M_W(\theta) = 1 + E(W)\theta + E(W^2)\frac{\theta^2}{2!} + E(W^3)\frac{\theta^3}{3!} + \dots$

we can equate terms to solve for $E(W^3)$

The θ^3 term in the series expansion for $M_W(\theta)$ is $\frac{(3\theta)^3}{3!} + 3\theta \cdot \theta^2 = \frac{45\theta^3}{3!}$

So $E(W^3)\frac{\theta^3}{3!} = 45\frac{\theta^3}{3!} \Rightarrow E(W^3) = 45$

Alternatively, you can use the fact that $E(W^3) = M_X'''(0)$.

(ii) $E(T^2) = E((e^W)^2) = E(e^{2W}) = M_W(2)$ by def'n of M_W

↑ since $T = e^W$ ↑ by algebra.

$= e^{3(2) + (2)^2}$

$= e^{10}$

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7 (a) (i) $T_{100} = X_1 + X_2 + \cdots + X_{100}$ where $X_1, X_2, \ldots, X_{100}$ are iid $\text{Exp}(4)$ (since $E(X) = 1/4$)

So $M_{T_{100}}(\theta) = M_{X_1}(\theta) M_{X_2}(\theta) \cdots M_{X_{100}}(\theta)$ since $X_1, \ldots X_{100}$ are indep.

$= (M_X(\theta))^{100}$ since $X_1, \ldots, X_{100}$ have the same distribution

$= \left(\frac{4}{4-\theta}\right)^{100}$ since $X \sim \text{Exp}(4)$ with $M_X(\theta) = \frac{4}{4-\theta}$

This is the mgf for $\Gamma(100, 4)$. Therefore, by the uniqueness of moment generating functions, $T_{100} \sim \Gamma(100, 4)$.

(ii) Central Limit Theorem: Suppose $X_1, X_2, \ldots$ are iid random variables with $E(X) = \mu$ and $\text{Var}(X) < \infty$. Then for any $-\infty < a < b < \infty$

$$\lim_{n \to \infty} P\left(a < \frac{X_1 + \cdots + X_n - n\mu}{\sqrt{n\sigma^2}} \leq b \right) = P(a < Z \leq b) \text{ where } Z \sim N(0,1).$$

(iii) $P(T_{100} > 28) = P\left(\frac{X_1 + \cdots + X_{100} - 100(1/4)}{\sqrt{100(1/4)^2}} > \frac{28 - 25}{\sqrt{100(1/4)^2}} \right)$

by CLT $\approx P\left(Z > \frac{28-25}{2.5}\right) = P(Z > 1.2)$ where $Z \sim N(0,1)$

$= 1 - P(Z \leq 1.2) = 1 - 0.8849$

$= 0.1151$

(b) (i) $Y \sim \text{Bin}(200, 0.4)$

(ii) $P(Y \leq 90) = P(Y \leq 90.5)$ by the continuity correction since Y is discrete

$= P\left(\frac{Y - 200(0.4)}{\sqrt{200(0.4)(0.6)}} \leq \frac{90.5 - 200(0.4)}{\sqrt{200(0.4)(0.6)}} \right)$

by the CLT $\approx P(Z \leq 1.52)$

$= 0.9357.$